Fractional Superstrings with Space-Time Critical Dimensions Four and Six

Source-backed arXiv sample

Abstract

We propose possible new string theories based on local world-sheet symmetries corresponding to extensions of the Virasoro algebra by fractional spin currents. They have critical central charges $c=6(K+8)/(K+2)$ and Minkowski space-time dimensions $D=2+16/K$ for $K\geq2$ an integer. We present evidence for their existence by constructing modular invariant partition functions and the massless particle spectra. The dimension 4 and 6 strings have space-time supersymmetry.

1 References And Citations

This sample cites source keys [1] and [2]. Section 1 and Equation 1 provide cross-reference coverage.

$$a^2 + b^2 = c^2 \tag{1}$$

References

[1] Source bibliography entry 1 extracted from arXiv metadata/source key ‘GSW‘.

[2] Source bibliography entry 2 extracted from arXiv metadata/source key ‘KMQ‘.

[3] Source bibliography entry 3 extracted from arXiv metadata/source key ‘BNY‘.

[4] Source bibliography entry 4 extracted from arXiv metadata/source key ‘ZFpara‘.